

Course

Data Science Research Methods

Reproducible Research Advanced Topics

Topics

- Advanced LaTeX
- Advanced R
- More on Knitr

More LaTeX

- child documents
- appendix
- bibliography
- (un)numbered lists

LaTeX – child documents

Main reasons for breaking up a document into subdocuments (“child document”):

- faster compilation of large documents
- working with several people on one report

The main (“master”) LaTeX document has entries like

`\include{chapterintroduction}` (no .tex extension!)

You may compile selected files with `\includeonly{file1,file2}`

Remarks:

- include automatically starts a new page (`\clearpage`)
- cannot be nested nor used in the preamble

LaTeX - appendix

Appendices are useful for including background information that you do not want to include in the main text since it would interrupt the reading flow.

`\appendix` % start the appendix

`\section{Your first appendix}\label{app:first}` % for articles

% for reports or books

`\chapter{Your first appendix}\label{app:first}`

The numbering restarts using capitals.

LaTeX - literature

Literature references are stored in .bib files (BibTeX)

```
@article{box1999statistics,  
  title={Statistics as a catalyst to learning by scientific method part {I}},  
  author={Box, G.E.P. and Liu, P.Y.T.},  
  journal={Journal of Quality Technology},  
  volume={31},  
  number={1},  
  pages={1},  
  year={1999}}
```

There are other formats like @book, @inproceedings,

Citations in this form:

```
\cite{Box99statistics}
```

The bibliography is put at the end of the document and is generated by the following commands:

```
\bibliographystyle{plain}
```

```
\bibliography{datascience} % for datascience.bib
```

Standard style: references are numbered.

Recommendation: use **natbib** package (add to preamble with option round) and style **plainnat**

Google Scholar



Scholar Settings

Search results

Languages

Library links

Account

Button

Collections

- ☒ Search articles (☒ include patents).
- ☐ Search case law.

Results per page

10

Google's default (10 results) provides the fastest results.

Where results open

- ☐ Open each selected result in a new browser window

Bibliography manager

- ☐ Don't show any citation import links.
- ☒ Show links to import citations into .



LateX – (un)numbered lists

Unnumbered lists (bullets):

```
\begin{itemize}
```

```
\item ...
```

```
\end{itemize}
```

Numbered lists: `\begin{enumerate}` etc. (may be nested)

You may change the appearance of the counters (preamble):

```
\renewcommand{\labelenumi}{\arabic(enumi)} etc.
```

There is a package `enumitem` that allows you to resume a numbered list (so remembering the counter):

```
\begin{enumerate}[resume]
```

More R

- functions
- apply
- aggregate
- condition by groups
- dates and times
- global options par, las, legend
- user information
- advanced graphics: ggplot
- regular expressions

R - functions

Functions are useful for automating operations and thus avoid repeating code.

```
myvar <- function(x) {sum((x-mean(x))^2/(length(x)-1)}
```

Note the use of a vectorized operation.

R - Loops

Loops in R are available through functions like **for**, **while**, **repeat** .

Try to avoid loops in R because:

- it may easily lead to errors with indexing variables
- it may be slow when dealing with large loops

The preferred way is to use the **apply** function or one its relatives (**sapply**,...) or one of the wrapper functions like **by** (to apply a function to a data frame split by rows through a factor).

R - Using Apply

Margins are the name in R to describe columns and/or rows.

1 = rows, 2 = columns, 1:2 = individual cells

`apply(x,1,mean)` # mean of the rows of x

`apply(x,2,mean)` # mean of the columns of x

`apply(x,1:2,function(x) {x/2})`

divide all elements of x by 2

There are useful variants like `sapply`

R - aggregation

aggregate data frame mtcars by cyl and vs,
by computing means of subgroups

access column names without specifying the data set
standard data set in R

attach(mtcars)

aggregate(mtcars, by=list(cyl,vs), FUN=mean, na.rm=TRUE)
note that “by” requires a list of factors

R - condition on groups

Conditioning is best performed by using factors and formula:

```
boxplot(bwt ~ smoke,data="CDHSbirthdata.txt")  
# plot bwt for every level of the factor smoke  
# ~ is a way to define a statistical model, more on this in  
next week
```

NB: Factors are internally represented as integers, and these are used if you would use `plot(x,y,...)`

R – dates and times

R allows to use the POSIXct format to store dates and times. Advantages:

- selection of dates
- computing with dates (since it is internally stored as seconds since 1-1-1970)
- fine-tuning of labels on plots

```
as.POSIXct(AEX$Date,format="%Y-%m-%d")
```

```
axis.POSIXct(1,at = seq(min(AEX$Date),  
max(AEX$Date),by = "6 months"),format="%b-%Y")
```

```
AEX[,AEX$Date]= &,]
```


R - regular expressions

Regular expressions increase search and replace capabilities. They are widely used, e.g. in cleaning data.

Examples:

```
grep("^J", month.name)
```

```
# find all months starting with J (positions)
```

```
grep("^J", month.name,value=T)
```

```
x <- c("R Tutorial","PHP Tutorial", "HTML Tutorial")
```

```
gsub("Tutorial","Examples",x)
```

R – global options

`par` can be used to set global options.

Example of graphical options:

```
par(las=1,pch=19,lwd=2)
```

`las` = orientation of labels on plots

`pch` = specify points in plots (recommended to use `pch=16` to have solid bullets instead of the circles – better visibility)

`lwd` = line width in plot (recommendation: `lwd = 2`)

Graph Types

dot chart: **dotchart**

strip chart: **stripchart**

histogram

- **hist** computes frequencies (and produces object)
- **plot** produces plot when applied to hist object

box and whisker plot: **boxplot**

bar chart: **barplot**

kernel density estimator:

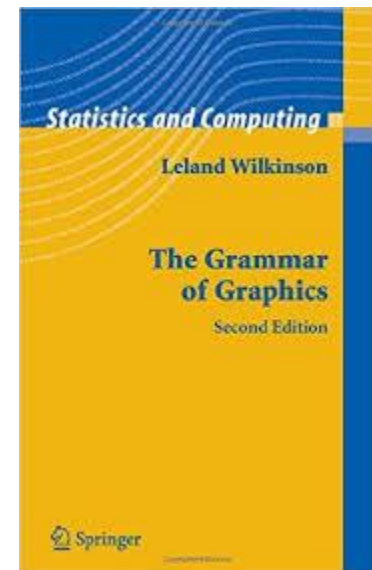
- **density** computes density object
- **plot** produces plot when applied to density object

R – ggplot2

Standard plot function in R works well for many plots, but has some drawbacks like cumbersome legends (does not scale well etc.).

Grammar of graphics was developed by Wilkinson to give a proper foundation of what graphs are.

Implementation in R by Wickham through the ggplot2 package.



Advice: use plot for simple plots, ggplot2 for advanced plots.

R - outline of grammar of graphics

Data

- Variables to be portrayed on the graph
- Variables are **mapped** to **aesthetics**, the perceivable features of the graph

Geoms

- Objects/shapes on the graph (e.g. points, bars)

Stats

- Statistical transformation, usually to summarize (e.g., mean, variance)

Scales

- Define which aesthetic values are mapped to variables (e.g., which colours are mapped to which values)

Coordinate Systems

- Define how data are mapped to the plane of the graphic (e.g., Cartesian)

Faceting

- Splits data into subsets to create multiple variations of the same graph (paneling)

R - qplot

A simple way to use ggplot is through **qplot** (q=quick)

```
qplot(mpg, wt, data = mtcars)
```

```
qplot(mpg, wt, data = mtcars, colour = cyl)
```

```
qplot(mpg, wt, data = mtcars, size = cyl)
```

```
qplot(mpg, wt, data = mtcars, facets = vs ~ am)
```

R – ggplot function

The ggplot function works with layers (indicated by +)

```
p <- ggplot(data=mtcars,aes(x=wt,y=mpg,color=cyl))
```

```
p + geom_line(shape=16)
```

change background to default and add title

```
p + geom_line(shape=16) + theme_classic() +  
labs(title="Cars")
```

box plot and density plot

```
p + geom_boxplot()
```

```
ggplot(data=mtcars,aes(x=wt,color=cyl)) + geom_density()
```

R – user info

Sys.info() returns several user related details, including:

- sysname
- login
- user
- machine

Useful when working in groups to e.g. set working directories

```
if (Sys.info()["user"] == "adibucch")  
{setwd ("C:/Users/adibucch/Documents")}
```


More knitr

- global options
- child documents
- caching
- Markdown
- LyX
- Word

Knitr - global options

```
<<globalsettings,echo=FALSE,results='hide'>>=  
opts_chunk$set(echo=F, results='hide', fig.width=6,  
fig.height=4, fig.align='center', message=F, warnings=F,  
fig.path="Figures/", plottune=T)
```

- It is good practice to put figures in a separate subfolder.
- Chunks remembers variables, but not global options

```
knit_hooks$set(plottune=function(before){  
if (before) par(las=1,pch=16,lwd=2)})  
@
```

Knitr - child documents

It is good practice (and useful when working in a group) to have one master file and a separate file for each chapter or section.

In LaTeX this is possible through `\include` .

In knitr, we need to include through chunks to ensure that all chunks in the child document work well:

1. make a main file , `report.nnw`

2. within the main file include chunks like

```
<<includechapter1,child="chapter1.rnw">>=
```

@

Note: the child files need no LaTeX preamble!

Knitr - caching

Loading a large data set or producing a complicated plot every time you compile your knitr document may be time consuming.

Caching is an option in which knitr remembers previous results so that they need not be executed again when they do not change.

```
<<yourchunk,caching=TRUE>>=
```

```
...
```

```
@
```

Knitr - LyX

LyX is a WYSIWYG editor for LaTeX available for Windows, Linux and Mac.



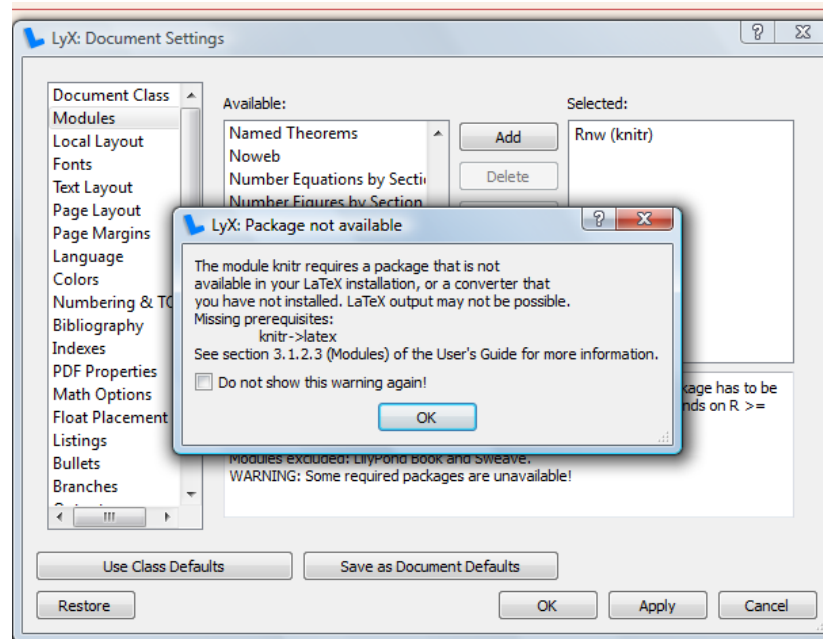
It has a menu for adding formulas etc but you can also use LaTeX commands for making formulas.

There are import and export facilities between the .lyx format and LaTeX / Word.

Knitr is supported by adding the module Rnw:

See <https://yihui.name/knitr/demo/lyx/>

Knitry– LyX configuration



This problem may be solved by setting an appropriate path, like you need to do for using LyX with Sweave:

<https://bitbucket.org/stat/lyx/downloads/sweave.pdf>

Knitr – R Markdown

Markdown is a simple markup language.

Advantages:

- export to Word and HTML possible
- it is possible to link with Python (see <https://pythonhosted.org/Markdown/>)

Supported by RStudio (library rmarkdown) with immediate rendering of mathematical formulas.

More information:

http://rmarkdown.rstudio.com/articles_docx.html

Summary

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- Knitr
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